

Boss Level

Classwork: Arrays, Loops & Everything You Know

Name:

Date:

Today you stop learning pieces and start building real programs. Every mission uses the SAME array — you'll just do something different with it each time. Predict before you run!

Before You Start — Setup (about 3 min)

Everything happens in your browser. Nothing to install!

1. Go to typescripqlang.org/play and clear out any sample code.
2. Type this at the top. EVERY mission today uses it — leave it there!

Type this code:

```
const scores: number[] = [85, 92, 78, 95, 88];
```

How many items are in scores? What is the LAST index?

REMEMBER

5 items means length is 5, but the indexes are 0, 1, 2, 3, 4. The last index is always length - 1.

Mission A — Print Each One (about 6 min)

Warm up the loop shape you'll use all day.

Type this code:

```
for (let i = 0; i < scores.length; i++) {  
  console.log(`Score ${i}: ${scores[i]}`);  
}
```

Write the FIRST line it printed:

Write the LAST line it printed:

Why does the first line say 0 and not 1?

Mission B — Total (about 6 min)

Move 1: add every item together. Notice where total is born!

Type this code:

```
let total: number = 0;

for (let i = 0; i < scores.length; i++) {
  total = total + scores[i];
}

console.log(`Total: ${total}`);
```

What was the total?

1. Break it: move the let total line INSIDE the loop and run again.

What did it print now, and why?

CONCEPT

Anything that has to REMEMBER across rounds must be created before the loop starts.

Mission C — Average (about 6 min)

Once you have the total, the average is one more line.

Type this code:

```
const average: number = total / scores.length;
console.log(`Average: ${average}`);
```

What was the average?

Why do we divide by scores.length instead of typing 5?

Mission D — Find the Biggest (about 10 min)

Move 2: hunt for the champion. Keep a variable holding the best one SO FAR.

Type this code:

```
let highest: number = scores[0];

for (let i = 0; i < scores.length; i++) {
  if (scores[i] > highest) {
    highest = scores[i];
  }
}

console.log(`Highest: ${highest}`);
```

What was the highest score?

Why does highest start at scores[0] instead of 0?

1. Challenge: change it to find the LOWEST score instead. What did you change?

Your answer:

Mission E — Count Them (about 8 min)

Move 3: same shape as Total — but add 1 instead of adding the value.

Type this code:

```
let aboveNinety: number = 0;

for (let i = 0; i < scores.length; i++) {
  if (scores[i] > 90) {
    aboveNinety = aboveNinety + 1;
  }
}

console.log(`${aboveNinety} scores were above 90`);
```

How many scores were above 90?

Change 90 to 80. How many now?

Mission F — Label Each One (about 6 min)

Put an if/else INSIDE the loop so every item gets judged.

Type this code:

```
for (let i = 0; i < scores.length; i++) {
  if (scores[i] >= 90) {
    console.log(`${scores[i]} - Great!`);
  } else {
    console.log(`${scores[i]} - Good`);
  }
}
```

How many said Great?

BIG IDEA

Notice the difference: Missions B-E printed ONE answer at the end. This one prints something for EVERY item. Where the `console.log` goes changes everything!

Mission G — Search for a Score (about 10 min)

Move 4: is a value in the list? This is where booleans finally get a real job.

Type this code:

```
const target: number = 78;
let found: boolean = false;

for (let i = 0; i < scores.length; i++) {
  if (scores[i] === target) {
    found = true;
  }
}

if (found) {
  console.log(`Yes! ${target} is in the list.`);
} else {
  console.log(`No, ${target} is not there.`);
}
```

What did it say?

1. Change target to 100 and run again.

What did it say this time?

Why do we start found at false instead of true?

Mission H — The AI Label Checker (about 8 min)

Remember the AI labels from your homework? Now let's CHECK that data with real code. This program uses everything you know.

Type this code:

```
const labels: string[] = ["tree", "unknown", "car", "tree", "unknown"];
let unknownCount: number = 0;

for (let i = 0; i < labels.length; i++) {
  if (labels[i] === "unknown") {
    unknownCount = unknownCount + 1;
  }
}

const percent: number = (unknownCount / labels.length) * 100;
console.log(`${unknownCount} of ${labels.length} are unknown (${percent}%)`);

if (percent > 30) {
  console.log("A human should review this data!");
} else {
  console.log("This data looks good enough to use.");
}
```

What percent were unknown?

Did it say a human should review it?

In your own words: why does it matter that a person checks AI data that has lots of 'unknown' labels?

Quick Quests (about 8 min)

Finished early? Check off as many as you can!

- ☐ Add two more scores to the array and re-run Missions B and D.
- ☐ Count how many scores are BELOW 85.
- ☐ Find the lowest score using the Biggest move in reverse.
- ☐ Make an array of your favourite foods and print each with its index.
- ☐ Search your food array for one item using the flag trick.
- ☐ Print only the scores that are even (use `% 2 === 0`).
- ☐ Add up only the scores above 85.
- ☐ STRETCH: find the highest score AND which index it was at.

Wrap-Up — Reflect (about 3 min)

Which of the four moves (Total, Biggest, Count, Search) was your favourite?

You've now learned types, variables, operators, if/else, loops, and arrays. What would you like to build with all of it?